

# JENSEN-SHANNON DIVERGENCE (JSD)

as a generalized, n-dimensional index of phonological separation



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## Main Claim

JSD provides the strongest correspondence to KDE-estimated distributional overlap across both traditional F1/F2 and high-dimensional MFCC acoustic spaces.

## 1. The Problem

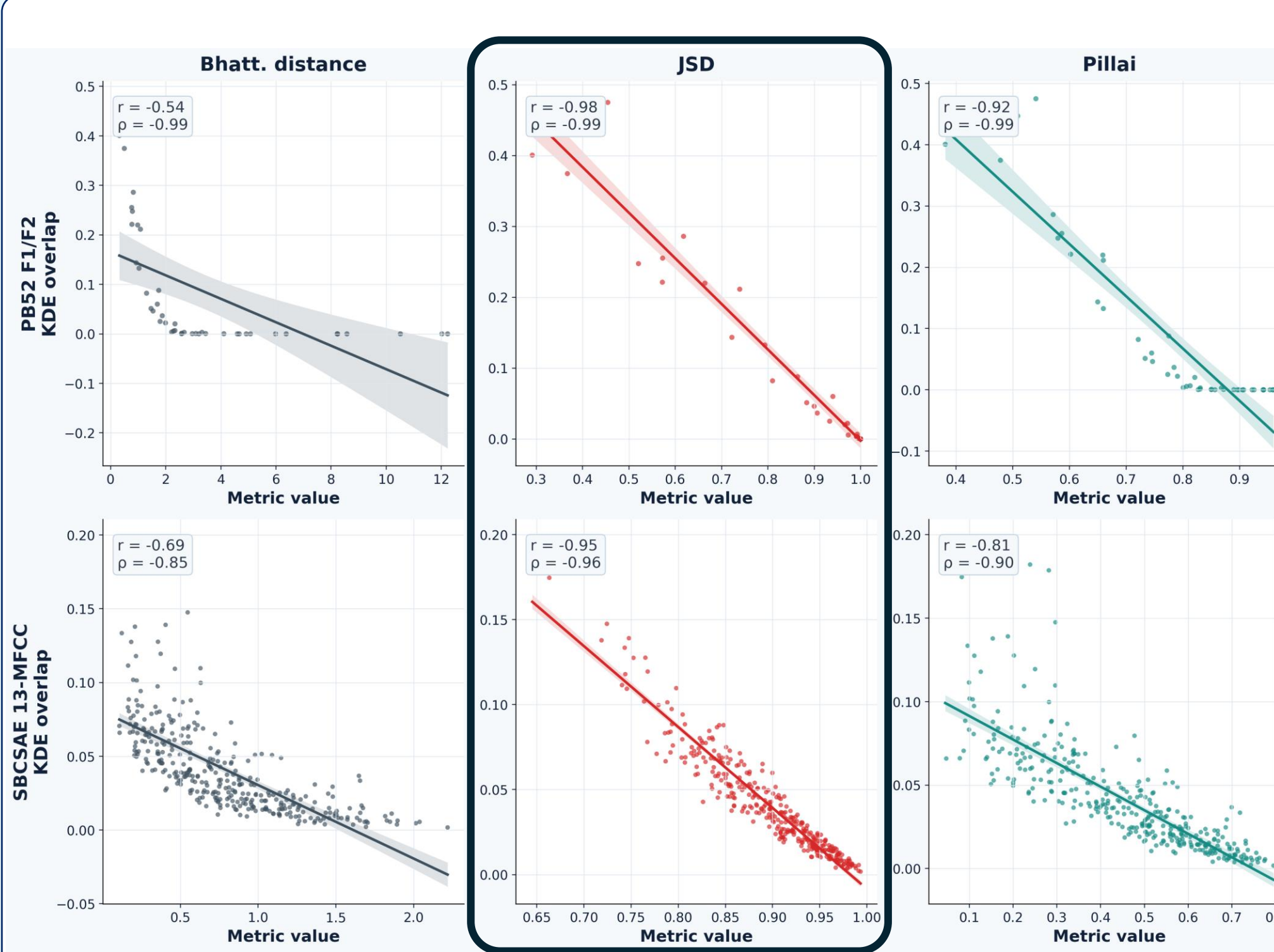
**Phonological contrast** is often quantified as the separation between acoustic category distributions. Commonly used measures each encode assumptions that become fragile in realistic phonological and sociophonetic data [5,7,8,10,11].

| Limitation                                           | Euc. | Bhatt. | Pillai | Maha. |
|------------------------------------------------------|------|--------|--------|-------|
| Reduces categories to centroid or mean separation    | ✗    | ✗      | ✗      | ✗     |
| Ignores distribution shape, skew, or multimodality   | ✗    | ✗      | ✗      | ✗     |
| Depends on covariance estimation                     | ✗    | ✗      | ✗      | ✗     |
| Requires stable/invertible covariance matrices       | ✗    | ✗      | ✗      | ✗     |
| Assumes linear/elliptical category structure         | ✗    | ✗      | ✗      | ✗     |
| Sensitive to unequal covariance across categories    | ✗    | ✗      | ✗      | ✗     |
| Depends on distributional assumptions                | ✗    | ✗      | ✗      | ✗     |
| Not bounded to a common [0,1] scale                  | ✗    | ✗      | ✗      | ✗     |
| Can miss gradient overlap between full distributions | ✗    | ✗      | ✗      | ✗     |

Euc.: Euclidean Distance; Bhatt.: Bhattacharyya Distance; Pillai: Pillai-Bartlett Trace; Maha.: Mahalanobis Distance

? Can a single, **model-agnostic measure (JSD)** compare full acoustic distributions across **low- and high-dimensional** feature spaces?

## 4. Main Result: JSD Best Matches Vowel Overlap



Top Row: PB52 (n = 1,520) using F1 and F2  
Bottom Row: SBCSAE (n = 13,067) using 13 MFCCs  
Each dot represents one pairwise comparison

### Overlap Benchmark

Nonparametric overlap between two category distributions via **kernel density estimation (KDE)**.

KDE overlap was used because it estimates distributions without assuming Gaussian form. JSD is also estimated using KDE in *phonJSD*<sup>[1]</sup> (§6). Five methods are compared on the MFCC data to test robustness to overlap estimator.

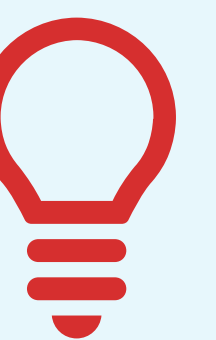
### Comparison Metrics

**Bhattacharyya Distance:** Covariance-based separation (ridge-regularized)

**JSD:** Distributional separation (KDE-based)

**Pillai-Bartlett Trace:** MANOVA-based separation

JSD recovers graded vowel separation more faithfully than the comparison metrics, especially in high-dimensional acoustic space.



## 2. What is JSD?

JSD<sup>[3]</sup> is an **information-theoretic measure** describing the similarity of two probability distributions, P, and Q:

$$JSD(P \parallel Q) = \frac{1}{2} \sum_{i=1}^n P(i) \log_2 \left( \frac{P(i)}{M(i)} \right) + \frac{1}{2} \sum_{i=1}^n Q(i) \log_2 \left( \frac{Q(i)}{M(i)} \right)$$

where  $M = \frac{1}{2}(P + Q)$  and  $n$  is the cardinality of the support of  $P$  and  $Q$ .

JSD is:

- **Symmetric** (direction doesn't matter)
- **Bounded** in [0,1] (0: Full overlap; 1: Full separation)
- **Generalizable** to n-dimensional feature spaces
- Easily **Convertible** to a metric ( $JSD_{\text{Dist}}(P, Q) = \sqrt{JSD(P, Q)}$ )
- **Model agnostic** (operates directly on the distributions)

Main degrees of freedom are **JSD estimation method**, **sample size**, how the probability distributions are **approximated**, and choice of **vowel measurement** (e.g., formants, MFCCs)

## 3. Comparison Data

### F1 / F2 Space Traditional vowel representation

Peterson & Barney<sup>[6]</sup> vowel data  
1,520 tokens  
10 vowels  
45 pairwise contrasts

### 13-MFCC Space Full spectral representation

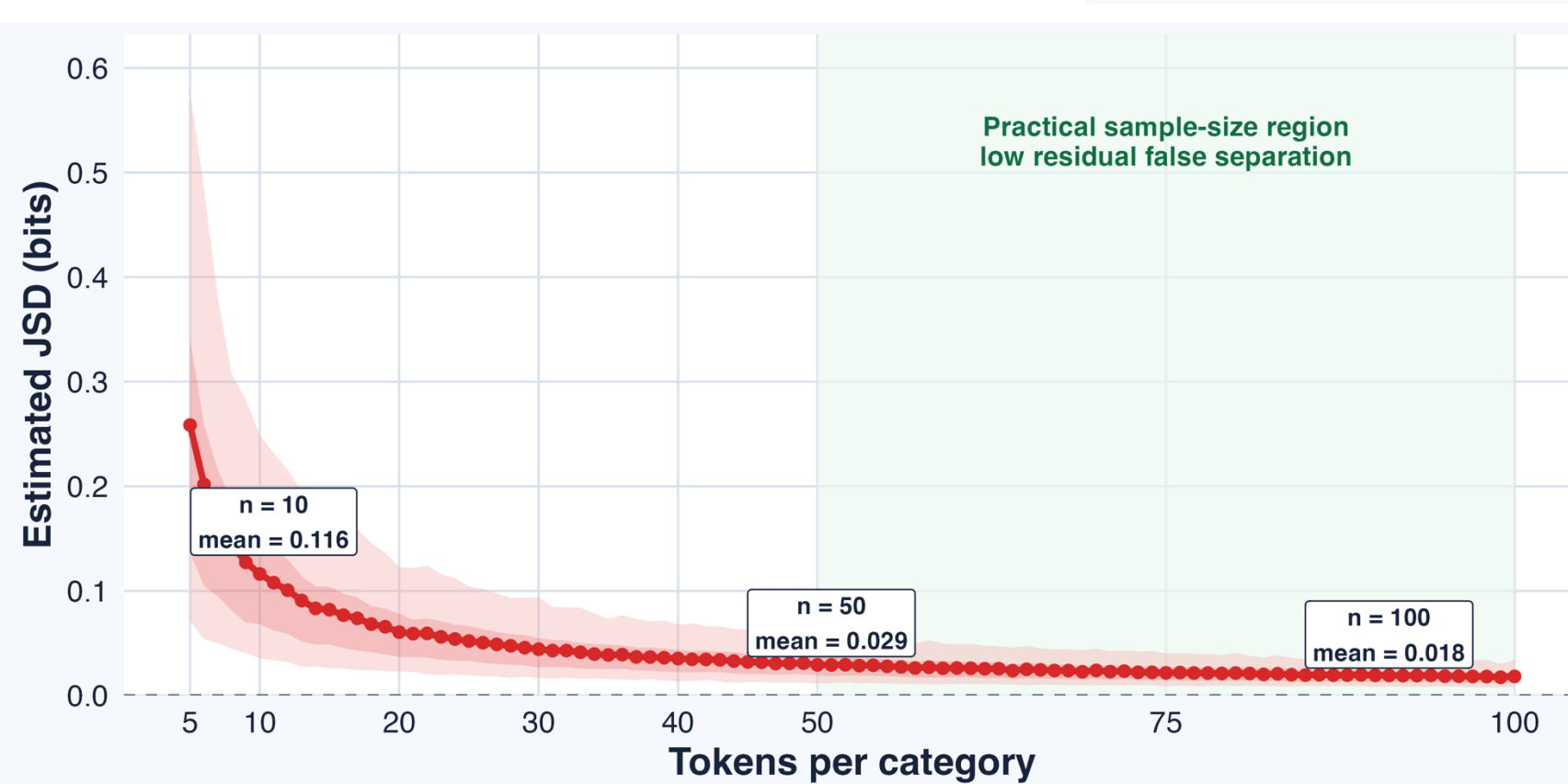
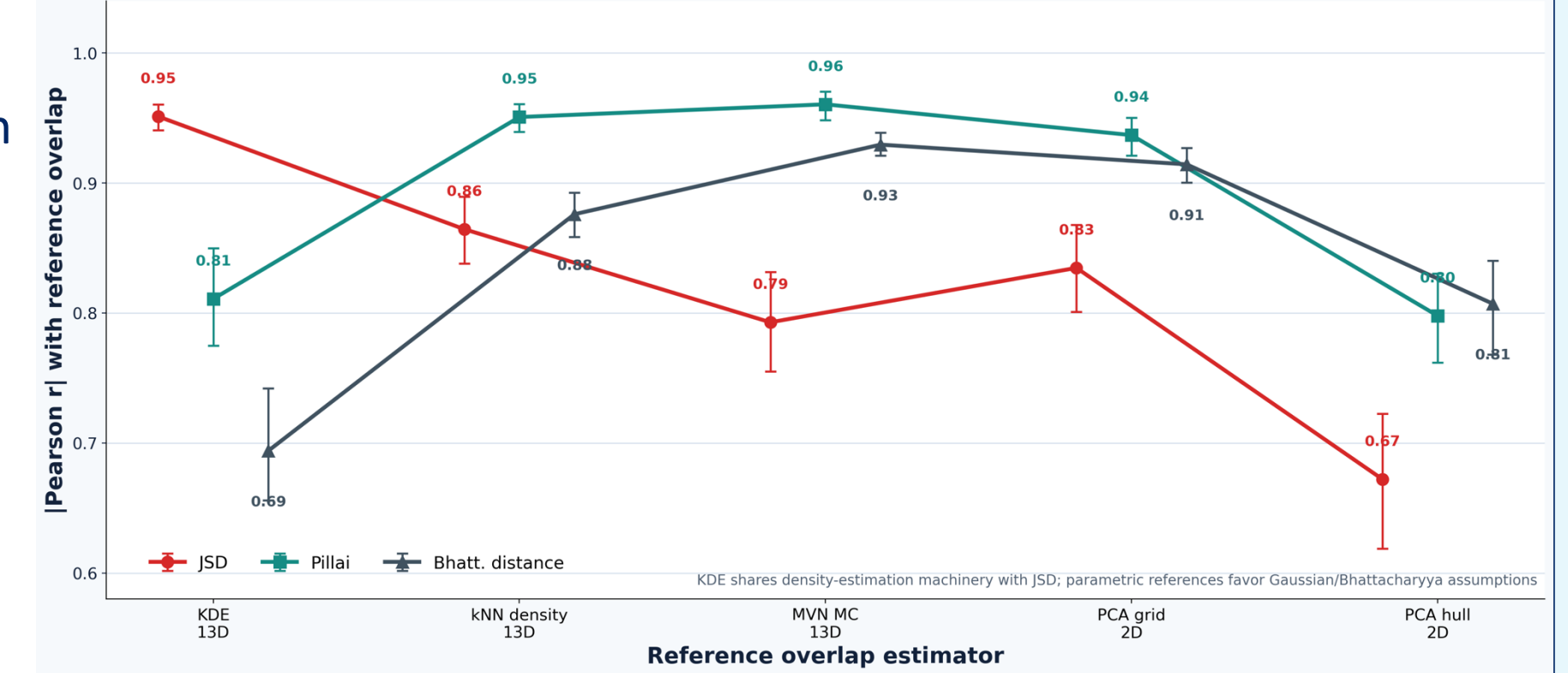
SBCSAE<sup>[2]</sup> force-aligned with MFA<sup>[4]</sup>  
200-500 tokens selected per vowel  
13,067 tokens after filtering/sampling (< 79k)  
27 vowels after filtering/sampling (< 32)  
351 pairwise contrasts  
KDE: diagonal Scott bandwidth, 200 points

## 5. Robustness Analyses

### Overlap Estimation

Category distributions were estimated with several alternative techniques. **JSD best predicts KDE-estimated overlap**; other criteria align more strongly with Pillai or Bhattacharyya metrics

KDE density: Nonparametric; widely used approach  
kNN density: Nonparametric; nearest-neighbor distances  
MVN Monte Carlo: Parametric; multivariate-normal models  
PCA-2D grid: Histogram overlap of two PCA dimensions  
PCA-2D convex hull: Geometric overlap of PCA (not density-based)

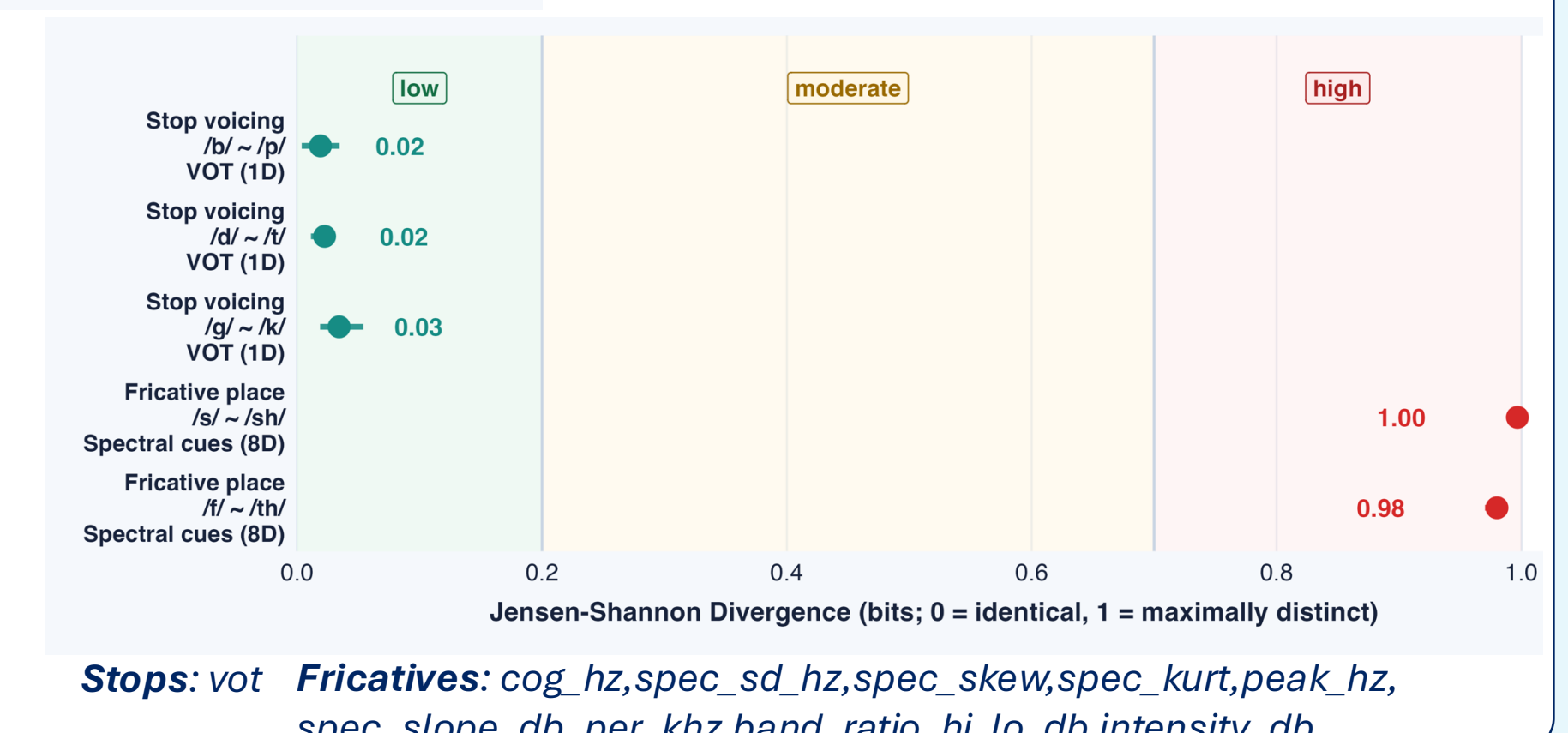


### Sample Size

Simulated load testing<sup>[9]</sup> was conducted for two hypothetical distributions with complete overlap at several sample sizes. Though JSD exhibits sensitivity to sample size, estimate **noise stabilizes after approximately 25 tokens per category** (mean: 0.05; 95% sim int: [0.02-0.12])

### Contrast

An exploratory analysis was conducted on stop voicing (VOT) and voiceless fricative place of articulation (8-d) in the SBCSAE data. Broadly, **JSD extends to voiceless fricative place of articulation** but not to VOT. This highlights what JSD measures: acoustic distributional separability in a specified feature space, **not perceptual cue weight** by itself.



## 6. Introducing phonJSD for R

**phonJSD<sup>[1]</sup>** is an open-source R package for information-theoretic measurement of phonological category separation.

### Current Features:

- Global and group-level JSD estimates
- Bootstrap **95% confidence intervals**
- **1-d and n-dimensional** feature spaces
- **Compare** with Pillai, Bhattacharyya, and KDE
- **Tidy** outputs for analysis and visualization



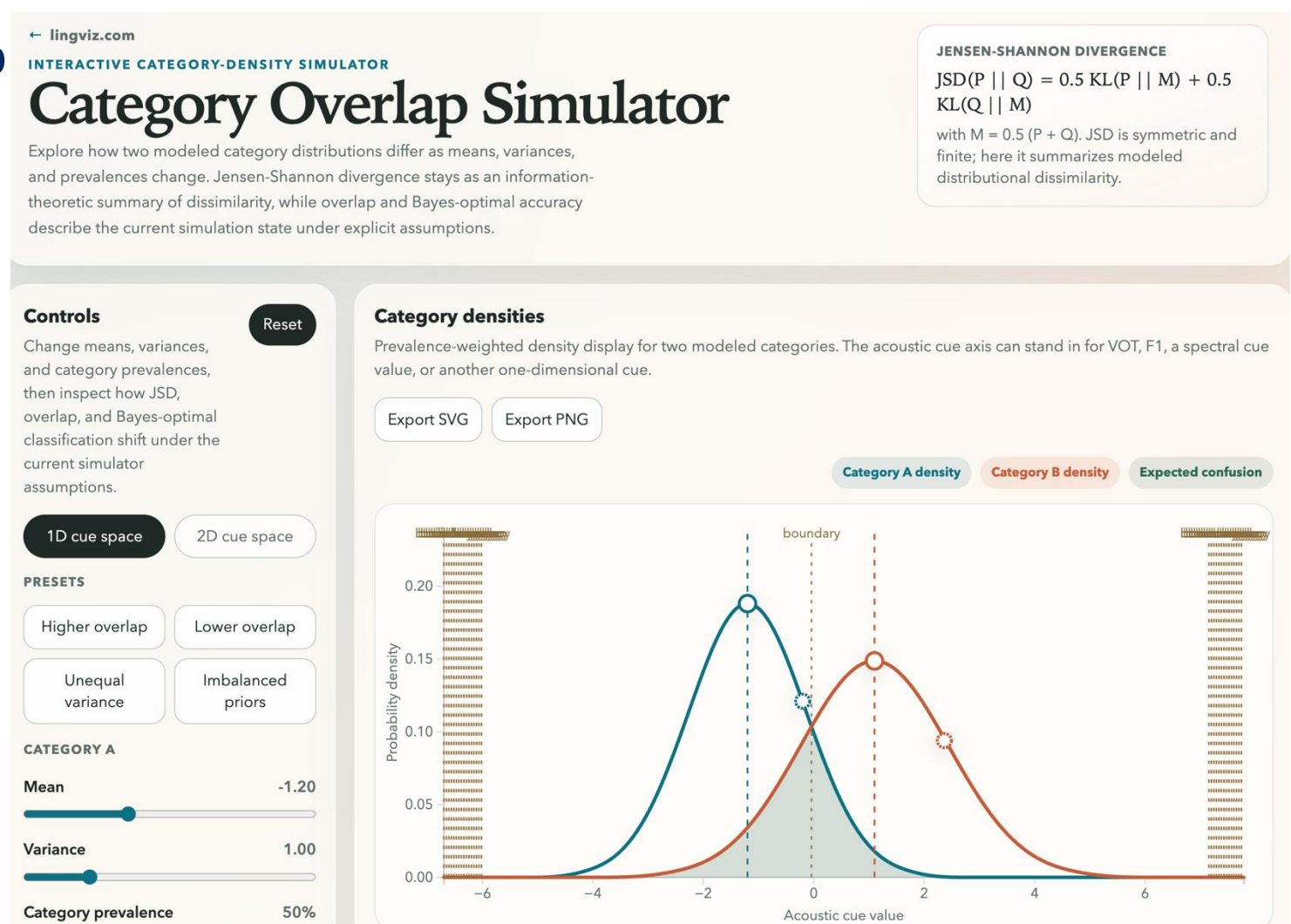
View it on Github!

Simulate Vowel Overlap on [lingviz.com!](https://lingviz.com/)

### Example Syntax

```
library(phonJSD)

estimate_jsd(
  data = vowels,
  features = c("F1",
    "F2"),
  category_col = "vowel",
  group_col = "speaker",
  do_boot = TRUE
)
```



## References

- [1] Berry, G.M. (2026). *phonJSD: 0.5.1 (0.5.1)*. Zenodo. <https://doi.org/10.5281/zenodo.20816586>; [2] Du Bois, J.W., Chafe, W.L., Meyer, C., Thompson, S.A., Englebreton, R., & Martey, N. (2000-2005). Santa Barbara corpus of spoken American English, Parts 1-4. Philadelphia: Linguistic Data Consortium.; [3] Lin, J. (1991). Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1), 145–151. <https://doi.org/10.1109/18.611115>; [4] McAuliffe, M., Socolof, M., Mihuc, S., Wagner, M., & Sonderegger, M. (2017). Montreal Forced Aligner: Trainable text-speech alignment using Kaldi. *Proceedings of Interspeech 2017*, 498–502.; [5] Nycz, J., & Hall-Lew, L. (2014). Best practices in measuring vowel merger. *Proceedings of Meetings on Acoustics*, 20(1), 060008.; [6] Peterson, G. E., & Barney, H. L. (1952). Control methods used in a study of the vowels. *The Journal of the Acoustical Society of America*, 24(2), 175–184. <https://doi.org/10.1121/1.1906875>; [7] Pillai, K.C.S. (1954). On some distribution problems in multivariate analysis. Special report to the Ford Foundation. *Institute of Statistics Mimeograph Series*, Number 88. Chapel Hill, NC: Institute of Statistics, University of North Carolina.; [8] Smith, I., Sonderegger, M., & The SPADE Consortium. (2024). Modelled multivariate overlap: A method for measuring vowel merger. arXiv. <https://arxiv.org/abs/2406.16319>; [9] Stanley, J. A., & Sneller, B. (2023). Sample size matters in calculating Pillai scores. *The Journal of the Acoustical Society of America*, 153(1), 54–67.; [10] Strelluf, C. (2018). Speaking from the Heartland: The Midland Vowel System of Kansas City. *Publication of the American Dialect Society* 103. Duke University Press: Durham, NC.; [11] Warren, P. (2018). Quality and quantity in New Zealand English vowel contrasts. *Journal of the International Phonetic Association*, 48(3), 305–330.